

SEP and MTOBJECTS Optuna Optimisation Process

What is optimised, how it is scored, and how run results are recorded

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Executive Summary

- There are now four optimisation routes: SEP Spike Gate, MTOBJECTS Spike Gate, MTOBJECTS toy-object, and SEP toy-object.
- Spike Gate optimisers optimise against narrow bar-major profile spikes, with a strong penalty for missed spike coverage and smaller penalties for unnecessary data loss.
- Toy-object optimisers inject known synthetic sources into real images and optimise recovery quality while penalising false-positive incremental masking. Current toy-object runs place those sources only inside the galaxy area being investigated, not across the full finite FITS footprint.

Toy-object injection region

For both SEP and MTOBJECTS toy-object optimisation, the valid injection region is the same deprojected, bar-aligned square cutout used by the normal visual reports and bar-major profiles. The x-axis is the deprojected bar-major axis and the y-axis is the deprojected bar-minor axis. The cutout radius follows the standard profile-view rule: $\max(3 \times \text{bar_sma}, 45 \text{ arcsec})$, limited by the available image boundary.

The optimiser samples toy centres only from finite pixels inside this investigated region. After the toy model is generated, the truth mask is formed from pixels above 8 percent of the toy peak and then truth-dilated if requested. The placement is accepted only if the full truth mask remains inside the investigated region.

Consequence: toy-object optimisation outputs produced before this rule was added on 2026-07-22 should be treated as superseded. They can overstate or misdirect performance because some injected objects may have been outside the area actually used for the galaxy/profile analysis.

- Each optimiser now appends run results to the shared workbook: D:\Dropbox\Public Documents\UCLAN\MSc Research\Remove foreground objects\documentation\Foreground Masking Optimisation Results.xlsx.
- The workbook records algorithm, method, run date/time, output paths, saved run configuration, trial parameters, best parameters, and per-trial results.

Programs Covered

Algorithm	Method	Program	Primary best-output
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			file
SEP	Spike Gate	Foreground Masking\optimise_sep_spike_gate_parameters.py	sep_spike_optimisation_best.json
MTOjects	Spike Gate	Foreground Masking\optimise_mtobjects_spike_gate_parameters.py	mtobjects_spike_optimisation_best.json
MTOjects	Toy Object	Foreground Masking\mtobjects_toy_object_parameter_optimisation.py	mtobjects_parameter_optimisation_best.json
SEP	Toy Object	Foreground Masking\sep_toy_object_parameter_optimisation.py	sep_toy_object_optimisation_best.json

Spike Gate Criteria

Spike Gate is a one-dimensional target generator. It does not produce the final foreground mask; it identifies profile samples where a foreground object appears as a narrow positive spike in the deprojected, bar-aligned major-axis intensity profile.

- The profile is measured from either the original or residual image, depending on the optimiser or GUI setting. The MTOjects Spike Gate optimiser exposes this as `--spike-gate-detect-on`.
- Candidate samples must be off-centre, using the configured centre-exclusion distance so the galaxy centre is not treated as a spike.
- A spike must be locally high relative to its neighbouring baseline, controlled by `spike-excess-fraction` and the inner/outer neighbour windows.
- The side-drop rule checks that the profile falls away on both sides, so broad galaxy structure is less likely to be selected.
- `spike-window-samples` expands each accepted spike sample so the scoring is tolerant of small radial alignment differences between the profile and two-dimensional mask.

Masking Processes

MTOjects

MTOjects is the two-dimensional segmentation engine for the MTOjects runs. The optimiser chooses global parameters and then the candidate mask is post-filtered before it is projected onto the science profile.

- `detect-on` selects whether MTOjects runs on the original image or the residual image.
- `move_factor` controls how far MTOjects moves objects during its internal watershed-style growth process.
- `min_distance` controls how close distinct detections can be before they merge or compete.
- `gaussian_fwhm` smooths the detection image before segmentation.
- `minarea` removes detections below a minimum pixel area.
- `dilation_radius` expands accepted masks after detection.
- `max_area` and `max_elongation` remove very large or highly elongated detections that are unlikely to be compact foreground objects.

SEP

SEP supplies a source-extraction segmentation mask for the SEP runs. The optimiser searches SEP detection and deblending parameters, then applies the same style of area, elongation, and dilation post-filtering used in the interactive SEP tools.

- `detect_thresh` sets the detection threshold above the local background.
- `minarea` sets the minimum connected area needed to keep a detection.
- `deblend_nthresh` and `deblend_cont` control SEP source splitting.
- `back_size` and `filter_size` control background estimation and smoothing.
- `dilation_radius` expands accepted masks.
- `max_area` and `max_elongation` reject broad or stretched detections after SEP extraction.

Optuna Optimisation and Scoring

All four optimisers use Optuna with a TPE sampler and a fixed seed. `initial-points` are exploratory startup trials; `max-iter` is added to `initial-points` to define the intended total trial count. Lower objective values are better.

Spike Gate Objective

For SEP Spike Gate and MTOjects Spike Gate, Optuna minimises:

```
objective = 12.0 * (1.0 - mean_spike_coverage)
           + 4.0 * mean_non_spike_profile_fraction
           + 2.0 * mean_masked_fraction
           + 1.5 * mean_profile_affected_fraction
           + 1.0 * mean_profile_change
```

The high missed-spike weight makes spike coverage the dominant goal. The remaining terms choose the least destructive mask among parameter sets that cover the target spikes.

Metric	Meaning
mean_spike_coverage	Fraction of Spike Gate target samples covered by the two-dimensional mask. Higher is better.
mean_non_spike_profile_fraction	Fraction of non-spike profile samples masked. Lower is better.
mean_masked_fraction	Fraction of image pixels masked. Lower is better.
mean_profile_affected_fraction	Fraction of profile samples touched by the mask. Lower is better.
mean_profile_change	Mean absolute profile change after masking/interpolation. Lower is better.

Toy-Object Score

SEP and MTOBJECTS now use separate Toy Objects objectives. Both score only incremental masking beyond the unaltered baseline and retain a 15% worst-image masking cap. Optuna minimises the negative feasible score; MTOBJECTS additionally applies explicit recovery infeasibility and final cross-fold release gates:

SEP: recovery = 0.45 mean recall + 0.20 mean F-score + 0.25 mean toy recall + 0.20 toy detection rate; loss = 0.35 mean masked fraction + 0.05 false-positive fraction. MTOBJECTS: recovery = 0.45 mean F-score + 0.35 mean toy recall + 0.20 toy detection rate; loss = 0.50 mean masked fraction + 0.10 false-positive fraction. MTOBJECTS trials are infeasible for an empty incremental mask, toy detection below 25%, or mean toy recall below 20%.

Metric	Meaning
mean_f_score	Balance of truth-pixel recall and precision for the incremental

	candidate mask.
mean_toy_recall	Mean fraction of injected truth pixels recovered.
toy_detection_rate	Fraction of injected objects with any recovered overlap.
false_positive_fraction	Incremental mask pixels outside injected truth, capped in the score penalty.

Run Commands

Run from C:\Users\gordo\Documents\Github\PythonScripts unless you pass fully qualified paths.

SEP Spike Gate initial run

```
python "Foreground Masking\optimise_sep_spike_gate_parameters.py" --max-images 20 --initial-points 16 --max-iter 64
```

SEP Spike Gate resume run

```
python "Foreground Masking\optimise_sep_spike_gate_parameters.py" --resume-output-dir "D:\Dropbox\Public Documents\UCLAN\MSc Research\Remove foreground objects\sep spike optimisation\YYYYMMDD_HHMMSS" --max-images 20 --initial-points 16 --max-iter 64
```

MTOBJECTS Spike Gate initial run

```
python "Foreground Masking\optimise_mtojects_spike_gate_parameters.py" --max-images 20 --initial-points 12 --max-iter 48 --spike-gate-detect-on residual --mtojects-detect-on original
```

MTOBJECTS Spike Gate resume run

```
python "Foreground Masking\optimise_mtojects_spike_gate_parameters.py" --resume-output-dir "D:\Dropbox\Public Documents\UCLAN\MSc Research\Remove foreground objects\mtojects spike optimisation\YYYYMMDD_HHMMSS" --max-images 20 --initial-points 12 --max-iter 48 --spike-gate-detect-on residual --mtojects-detect-on original
```

MTOBJECTS toy-object run

```
python "Foreground Masking\mtojects_toy_object_parameter_optimisation.py" --max-images 20 --toys-per-image 6 --initial-points 8 --max-iter 32 --mtojects-detect-on original
```

SEP toy-object run

```
python "Foreground Masking\sep_toy_object_parameter_optimisation.py" --max-images 20 --toys-per-image 6 --initial-points 8 --max-iter 32 --detect-on residual
```

Outputs and Workbook Logging

- Each run creates a timestamped output directory under the relevant optimisation folder unless `--output-dir` or `--resume-output-dir` is supplied.
- Each optimiser writes a summary CSV with one row per trial and a details CSV with per-galaxy or per-toy-object metrics.
- Each optimiser writes a best JSON containing the current best objective/score and the associated parameter set.
- Spike Gate runs also write a cases CSV; toy-object runs write a toys CSV catalogue.
- At the end of a non-prepare-only run, the optimiser appends its summary rows to the shared XLSX workbook.

Worksheet	Rows currently backfilled
MTOBJECTS Spike Gate	120 trial rows from two completed runs
SEP Spike Gate	1 trial row from the run currently present
MTOBJECTS Toy Object	Corrected runs append automatically; pre-2026-07-22 whole-FITS toy outputs are superseded.
SEP Toy Object	Corrected runs append automatically; use the 2026-07-22 investigated-area run or later.

The workbook columns include algorithm, method, run_datetime, run_dir, source file paths, best objective/score, best_params_json, config_* run settings, trial parameter columns, and trial result columns.

Important Limitations

- Spike Gate only rewards removal of foreground structure represented as selected profile spikes. Real foreground objects outside those samples are only indirectly handled by the masking algorithm and data-loss penalties.
- Toy-object optimisation answers a different question: recovery of known injected compact objects inside the investigated galaxy cutout. It is useful for general object-recovery behaviour but is not a direct spike-removal objective.
- Changing detect-on from residual to original can change the optimum because the segmentation input image changes.

- The SEP and MTOjects toy-object optimisers now enforce the investigated-area injection rule. Any older toy-object run created before 2026-07-22 should be rerun before using its best parameters as production defaults.

Systematic documentation review - 16 August 2026

Review classification: Updated - current Toy Objects optimisation process

Updates made: The shared pre-follow-up Toy Objects score was replaced by separate SEP and MTOjects objective definitions. Four-fold candidate evaluation, recovery feasibility and release-gate behaviour were added.

Current interpretation: SEP remains the primary result. MTOjects follow-up optimisation is valid but should be used as a review-controlled comparison because recovery is accompanied by substantial collateral masking.

Authoritative combined outcome: SEP and MTOjects Toy Objects Optimisation Summary and Conclusions.docx. Full-precision results: Foreground Masking Optimisation Results.xlsx.

Current four-fold design: 40 validated galaxies; four rotations of 30 training and 10 held out; 40 trials per fold (8 startup + 32 TPE); candidates compared on the same common-40 injection set. Selected winners: SEP fold 4 and MTOjects fold 3.